University of Florida, Department of Mathematics

Analysis First-Year Exam, January 2018, Part 2

Answer FOUR questions in detail. State carefully any results used without proof.

1. Let $f : [0,1] \to \mathbb{R}$ be Riemann-integrable. Prove that if the Riemann integral $\int_0^1 f$ is nonzero then there exist $\delta > 0$ and a nonempty open interval $I \subseteq [0,1]$ such that $|f| \geq \delta$ throughout I.

2. Let $f_n(t) = t^n(1-t)^n$ whenever $0 \leq t \leq 1$ and n is a positive integer. Does the sequence $(f_n : n > 0)$ converge on $[0,1]$ pointwise? Uniformly? Justify.

3. Let $f : [0,1] \to \mathbb{R}$ be continuous. Assume that $\int_0^1 f(t)e^{-nt}\,dt = 0$ whenever n is a non-negative integer. Does it follow that f is identically zero? Does the answer change if the exponent $-nt$ is replaced by $+nt$? By $2nt$? Justify.

4. Given that $(f_n : n \geq 0)$ is a sequence of measurable real-valued functions, explain why each of the following sets is (or is not) measurable:

(i) $X = \{\omega : f_n(\omega) \to \infty \text{ as } n \to \infty\}$;

(ii) $Y = \{\omega : f_n(\omega)^2 < f_n(\omega) \text{ for each } n \geq 0\}$;

(iii) $Z = \left\{\omega : \sum_{n=0}^{\infty} f_n(\omega) \text{ converges absolutely}\right\}$.

5. Let $(f_n : n \geq 1)$ be a sequence of continuous functions from $[0,1]$ to $[0,1]$ and assume that this sequence converges to f pointwise. True or false (proof or counterexample)?

(i) f is Riemann-integrable and (Riemann integrals) $\int_0^1 f_n \to \int_0^1 f$;

(ii) f is Lebesgue-integrable and (Lebesgue integrals) $\int_0^1 f_n \to \int_0^1 f$.